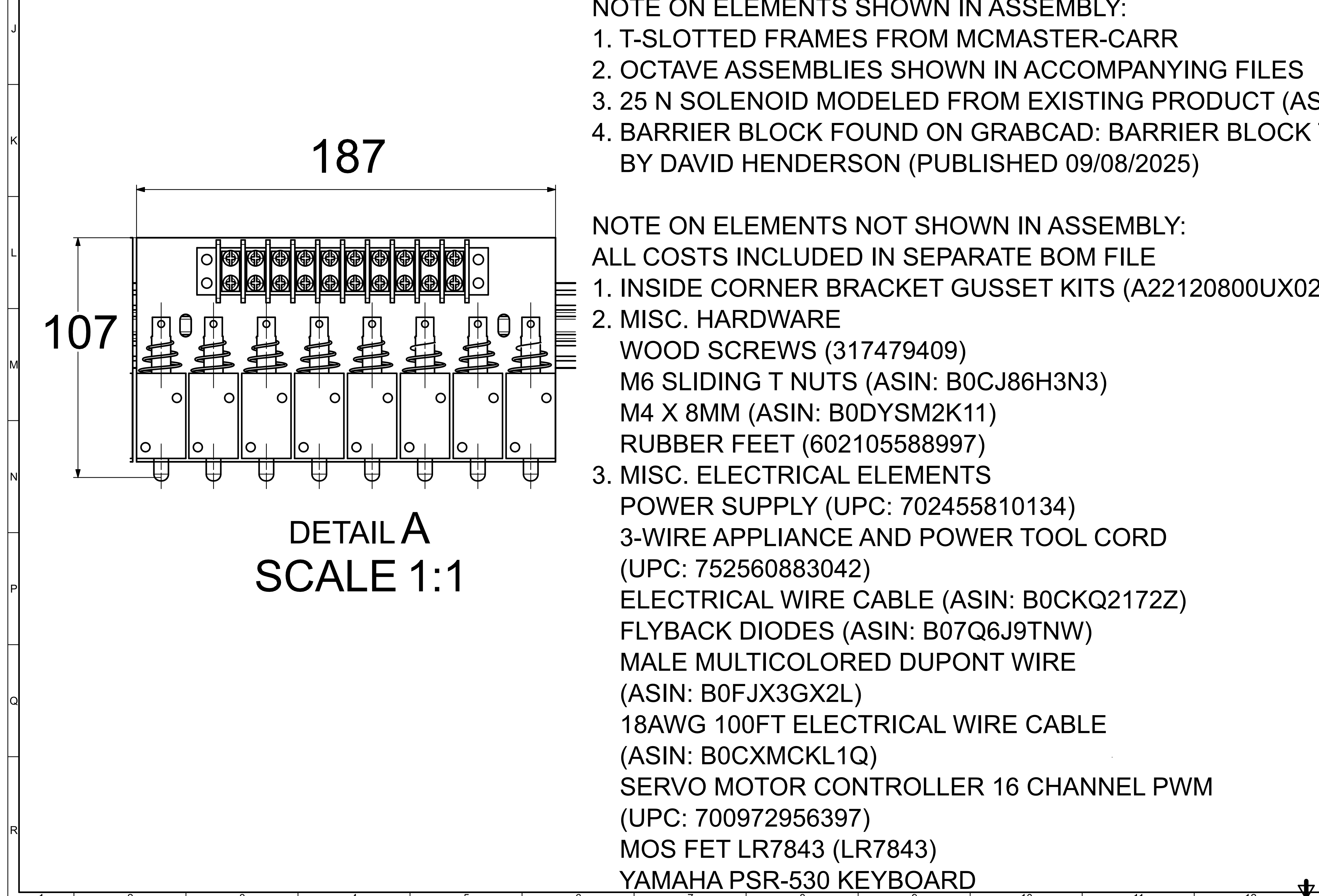
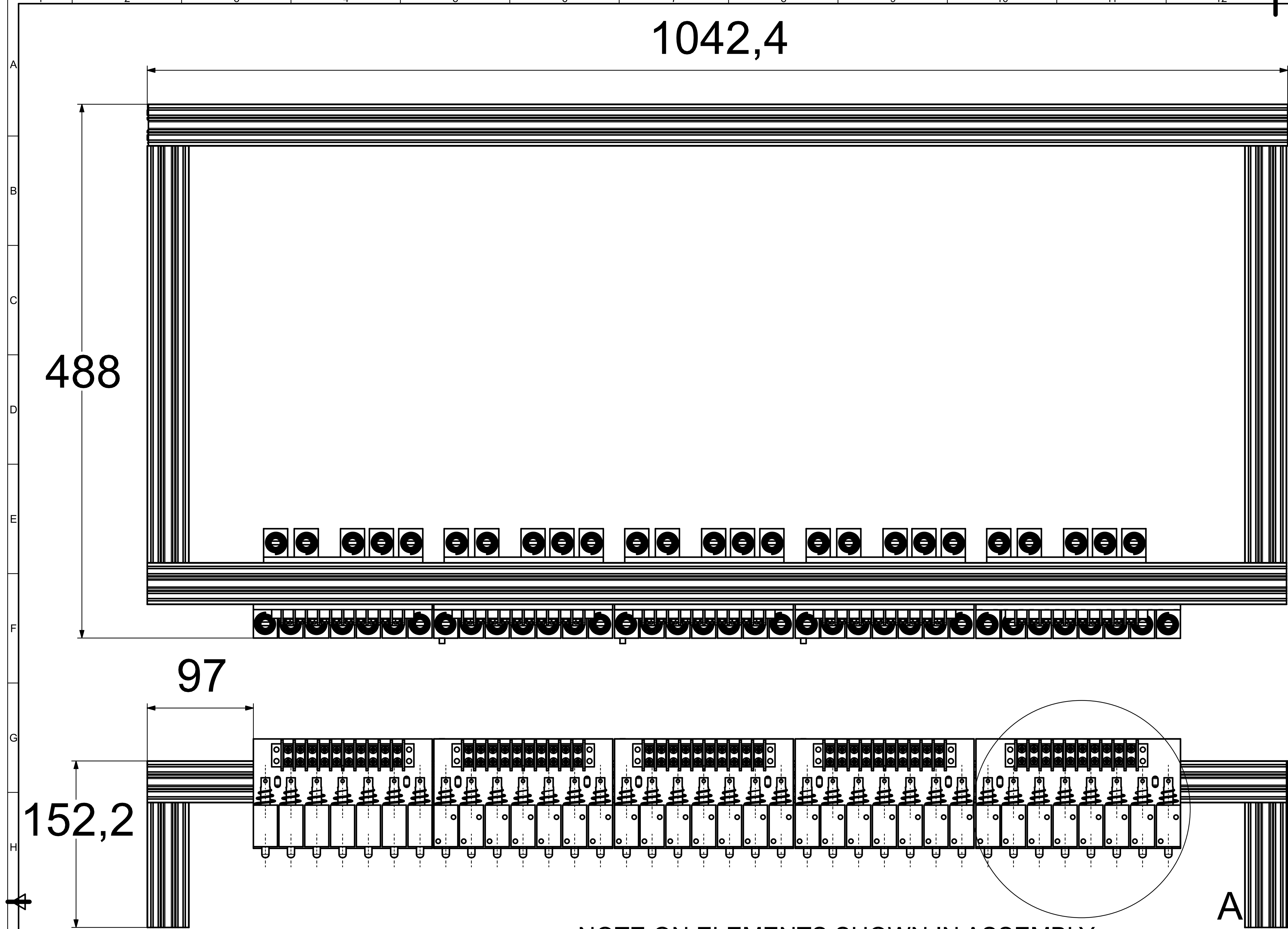




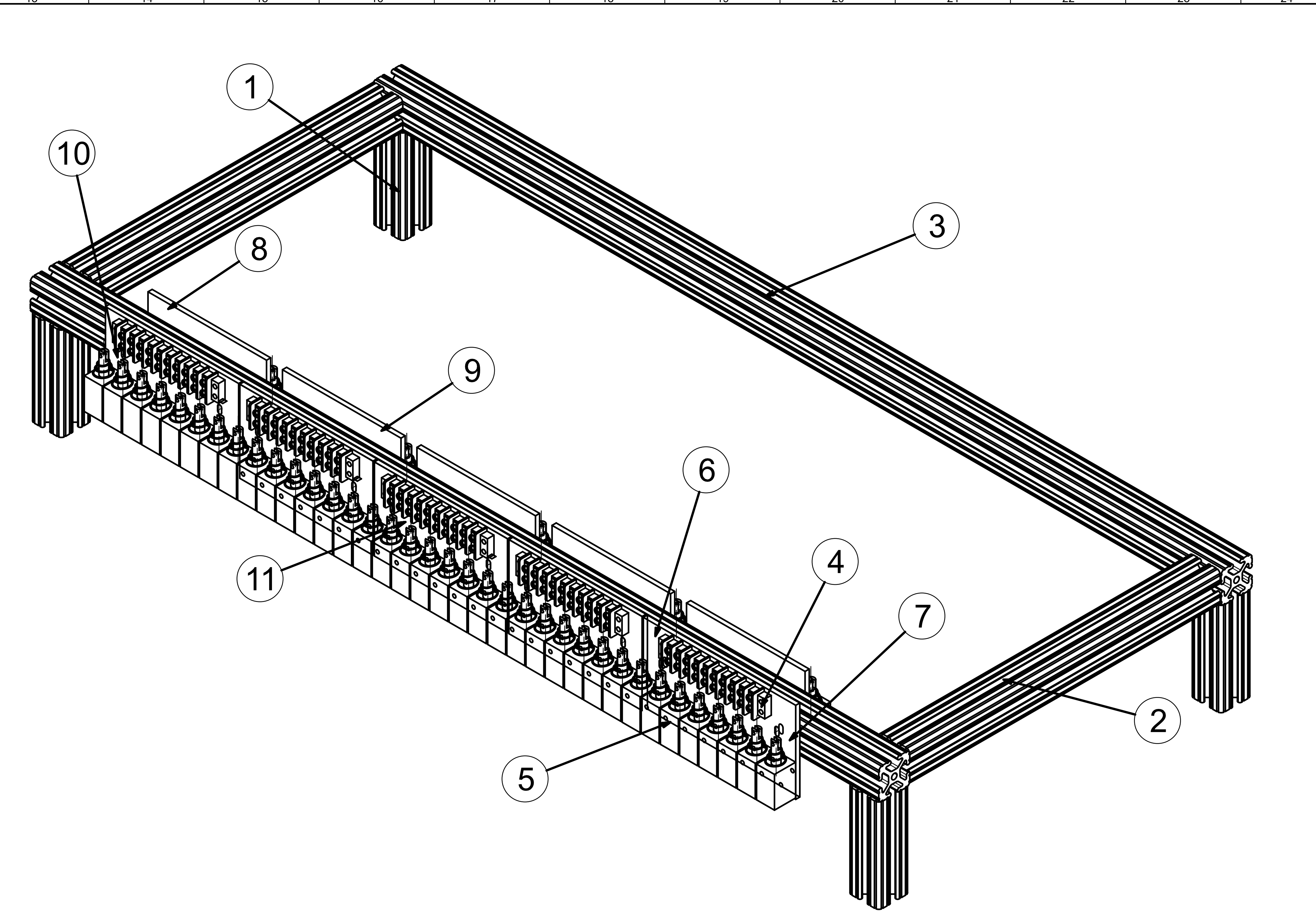
NOTE ON ELEMENTS SHOWN IN ASSEMBLY:

- 1. T-SLOTTED FRAMES FROM MCMMASTER-CARR
- 2. OCTAVE ASSEMBLIES SHOWN IN ACCOMPANYING FILES
- 3. 25 N SOLENOID MODELED FROM EXISTING PRODUCT (ASIN: B07Z4JQPN)
- 4. BARRIER BLOCK FOUND ON GRABCAD: BARRIER BLOCK TE 1-1546670-0 BY DAVID HENDERSON (PUBLISHED 09/08/2025)

NOTE ON ELEMENTS NOT SHOWN IN ASSEMBLY:

ALL COSTS INCLUDED IN SEPARATE BOM FILE

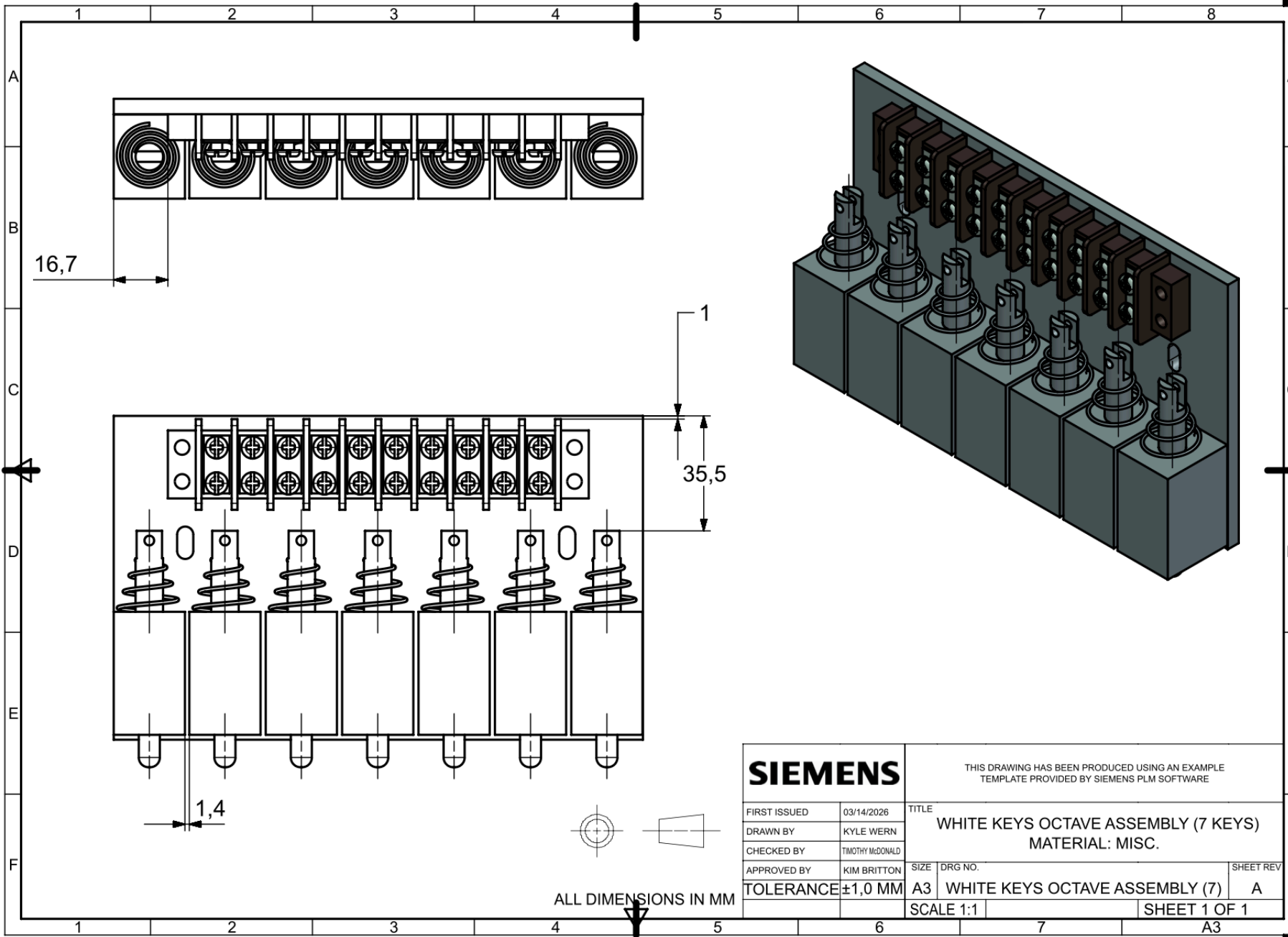
- 1. INSIDE CORNER BRACKET GUSSET KITS (A22120800UX0241)
- 2. MISC. HARDWARE
 - WOOD SCREWS (317479409)
 - M6 SLIDING T NUTS (ASIN: B0CJ86H3N3)
 - M4 X 8MM (ASIN: B0DYSM2K11)
 - RUBBER FEET (602105588997)
- 3. MISC. ELECTRICAL ELEMENTS
 - POWER SUPPLY (UPC: 702455810134)
 - 3-WIRE APPLIANCE AND POWER TOOL CORD (UPC: 752560883042)
 - ELECTRICAL WIRE CABLE (ASIN: B0CKQ2172Z)
 - FLYBACK DIODES (ASIN: B07Q6J9TNW)
 - MALE MULTICOLORED DUPONT WIRE (ASIN: B0FJX3GX2L)
 - 18AWG 100FT ELECTRICAL WIRE CABLE (ASIN: B0CXMCKL1Q)
 - SERVO MOTOR CONTROLLER 16 CHANNEL PWM (UPC: 700972956397)
 - MOS FET LR7843 (LR7843)
 - YAMAHA PSR-530 KEYBOARD



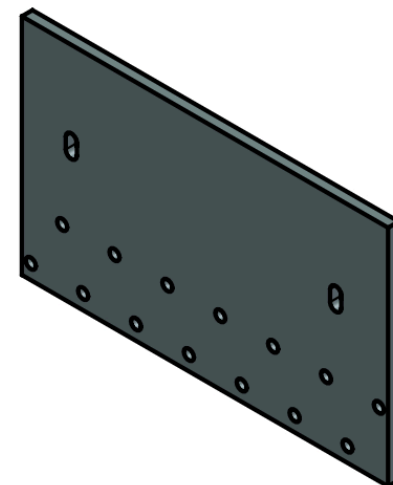
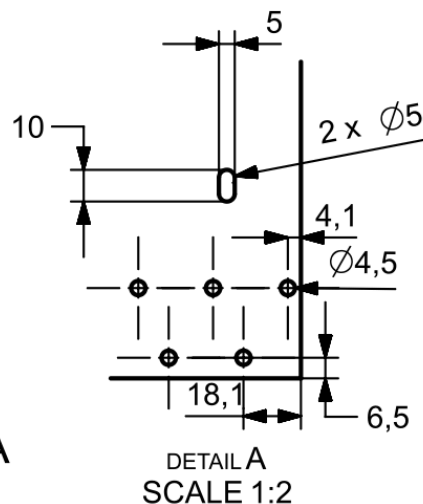
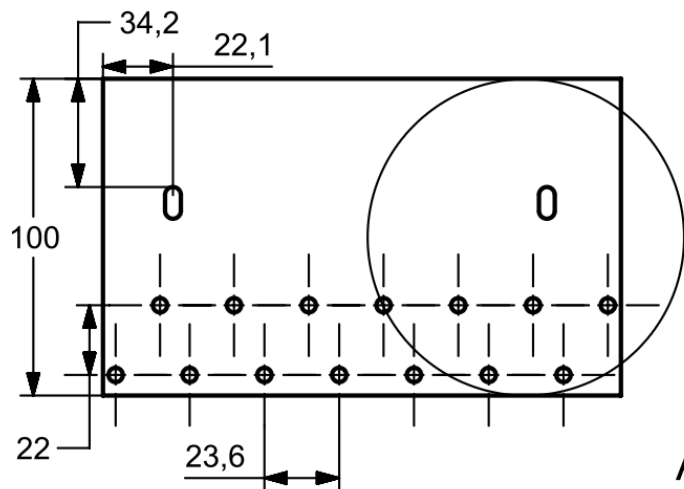
[N/A]	[MOSFET MOUNTS]	[4]	4.00
[N/A]	[PEDAL HOUSING ASSEMBLY]	[1]	1.00
PC NO	PART NAME	QTY	USD.

[N/A]	[PCA9685 MOUNTS]	[16]	16.00
11	[WHITE KEYS OCTAVE ASSEMBLY (7 KEYS)]	3	3.00
10	[WHITE KEYS SOLENOID MOUNTING (7 KEYS) BACKPLATE]	4	4.00
9	[BLACK KEYS OCTAVE ASSEMBLY (5 KEYS)]	4	4.00
8	[BLACK KEYS SOLENOID MOUNTING (5 KEYS) BACKPLATE]	5	5.00
7	[WHITE KEYS OCTAVE ASSEMBLY (8 KEYS)]	1	1.00
6	[WHITE KEYS SOLENOID MOUNTING (8 KEYS) BACKPLATE]	1	1.00
5	[25 N SOLENOID]	[62]	210.8
4	[BARRIER BLOCK]	5	27.78
3	[McMASTER-CARR 1041 MM CUT]	2	63.27
2	[McMASTER-CARR 381 MM CUT]	2	23.15
1	[McMASTER-CARR 114 MM CUT]	4	17.50
PC NO	PART NAME	QTY	USD.

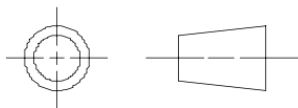
SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	06/02/2026			
DRAWN BY	K. BRITTON	TITLE MUSICAL BOT ENCORE FULL ASSEMBLY MATERIAL: MISC.		
CHECKED BY	T. McDONALD			
APPROVED BY	K. WERN	SIZE DRG NO. SHEET REV A0 FINAL MUSICAL BOT ENCORE FULL ASSEMBLY A		
		SCALE 1:2	SHEET 1 OF 1	



SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	03/14/2026	TITLE WHITE KEYS OCTAVE ASSEMBLY (7 KEYS) MATERIAL: MISC.		
DRAWN BY	KYLE WERN			
CHECKED BY	TIMOTHY McDONALD	SIZE	DRG NO.	SHEET REV
APPROVED BY	KIM BRITTON	A3	WHITE KEYS OCTAVE ASSEMBLY (7)	A
TOLERANCE	±1,0 MM	SCALE 1:1	SHEET 1 OF 1	



NOTE:
 2 X THRU SLOT 10 LG X 10 W X R5
 14 X 4,5 MM THRU HOLES
 1 MM TOLERANCE APPLIED
 FOR PETG FILAMENT PRINTING



ALL DIMENSIONS IN MM

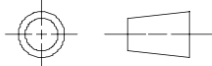
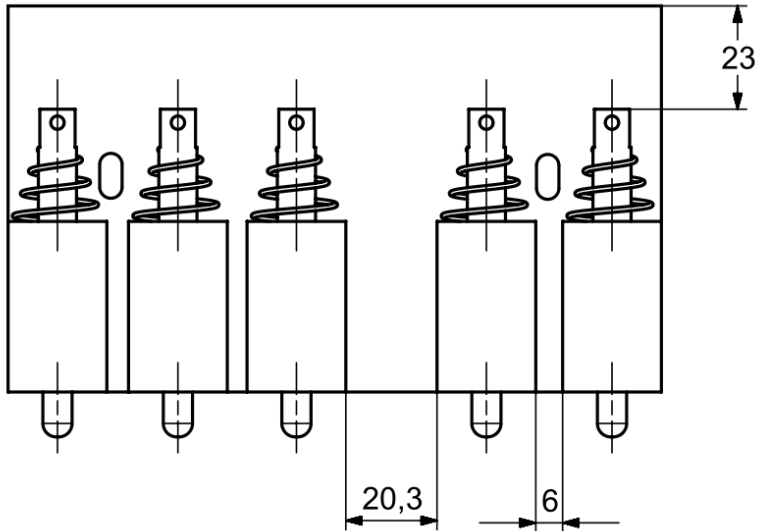
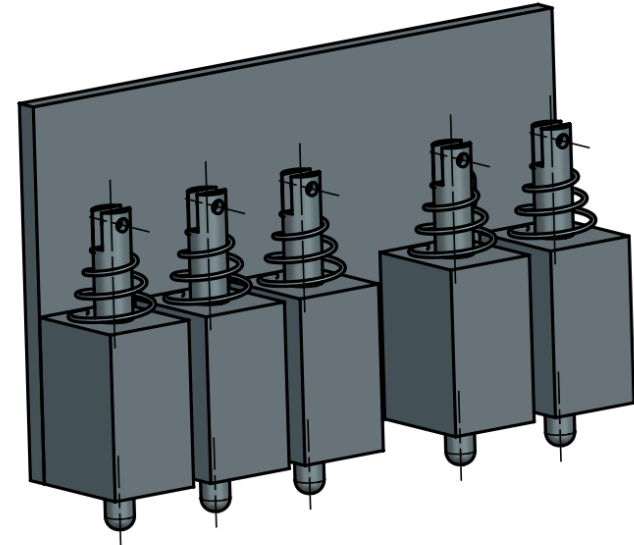
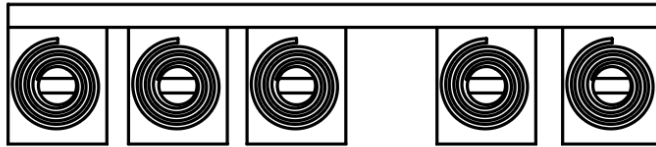
SIEMENS

THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
 TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE

FIRST ISSUED	03/14/2026
DRAWN BY	KYLE WERN
CHECKED BY	TIMOTHY McDONALD
APPROVED BY	KIM BRITTON
TOLERANCE	±1,0 MM

TITLE			WHITE KEYS SOLENOID MOUNTING (7 KEYS) BACKPLATE MATERIAL: PETG FILAMENT		
SIZE	DRG NO.				SHEET REV
A4	WHITE KEYS SOLENOID MOUNTING (7 KEYS) BACKPLATE				A
SCALE 1:1				SHEET 1 OF 1	

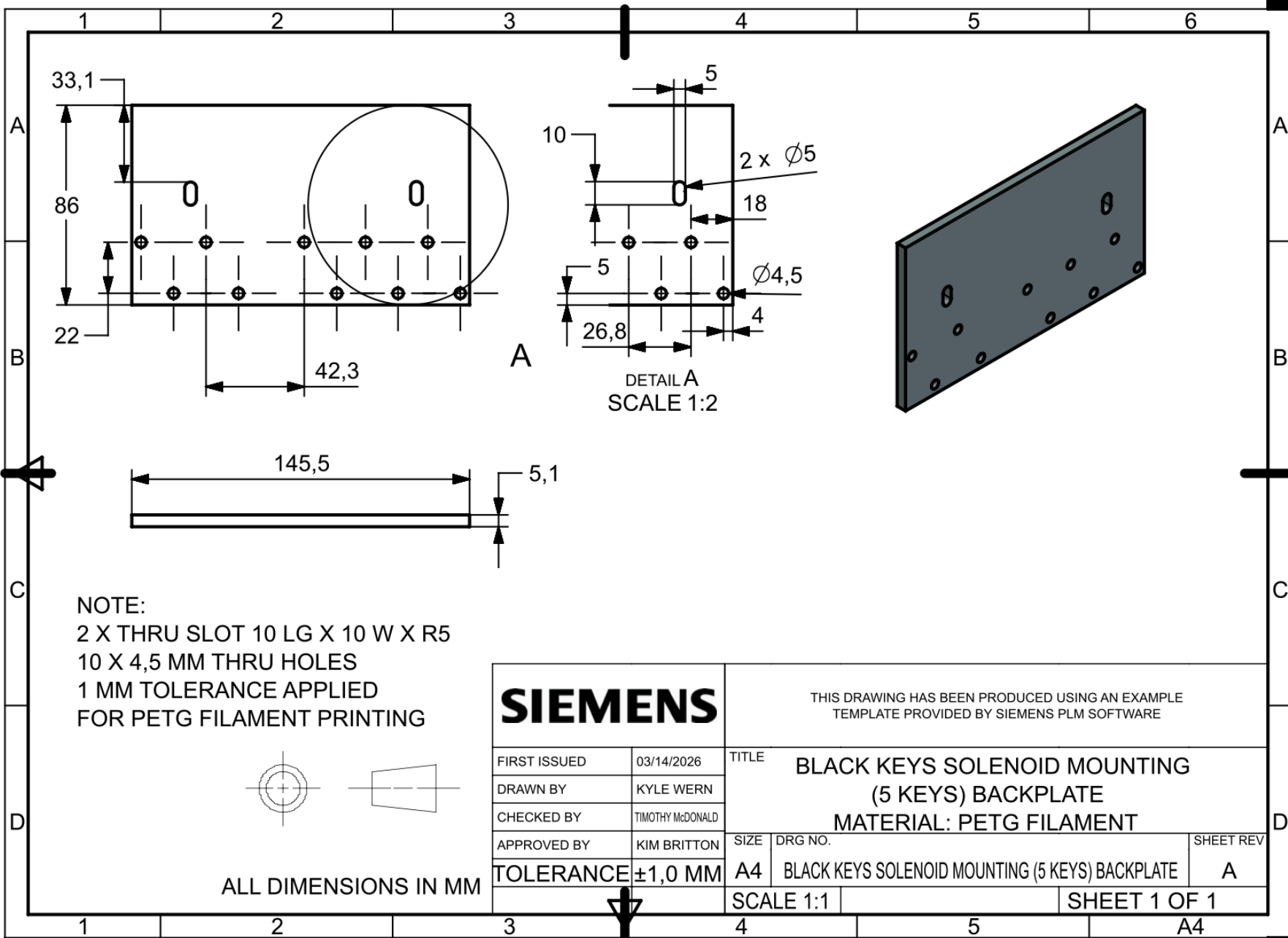
A4

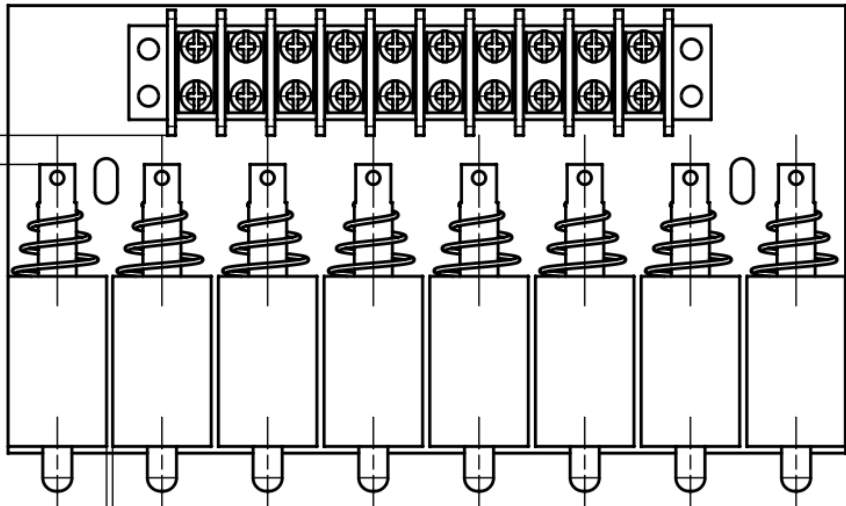
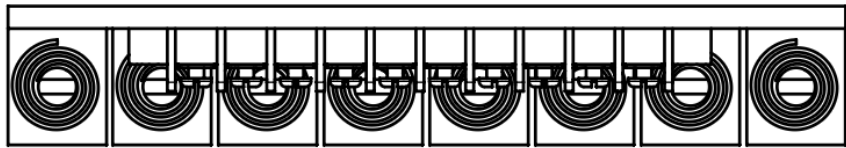


ALL DIMENSIONS IN MM

SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE	
FIRST ISSUED	03/14/2026	TITLE	
DRAWN BY	KYLE WERN	BLACK KEYS OCTAVE ASSEMBLY (5 KEYS)	
CHECKED BY	TIMOTHY McDONALD	MATERIAL: MISC	
APPROVED BY	KIM BRITTON	SIZE	DRG NO.
TOLERANCE $\pm 1,0$ MM		A3	BLACK KEYS OCTAVE ASSEMBLY
		SCALE 1:1	SHEET 1 OF 1

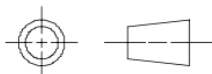
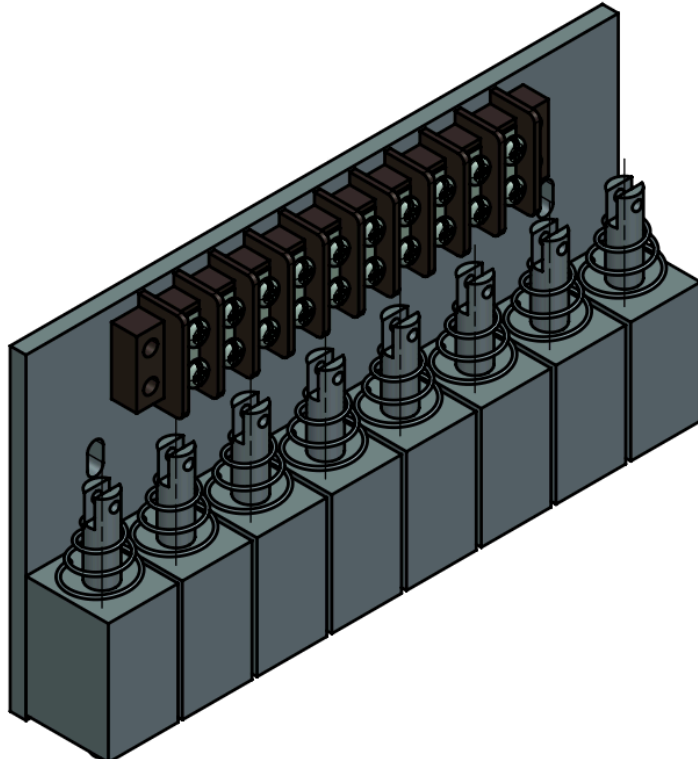
A3





6,5

1,4



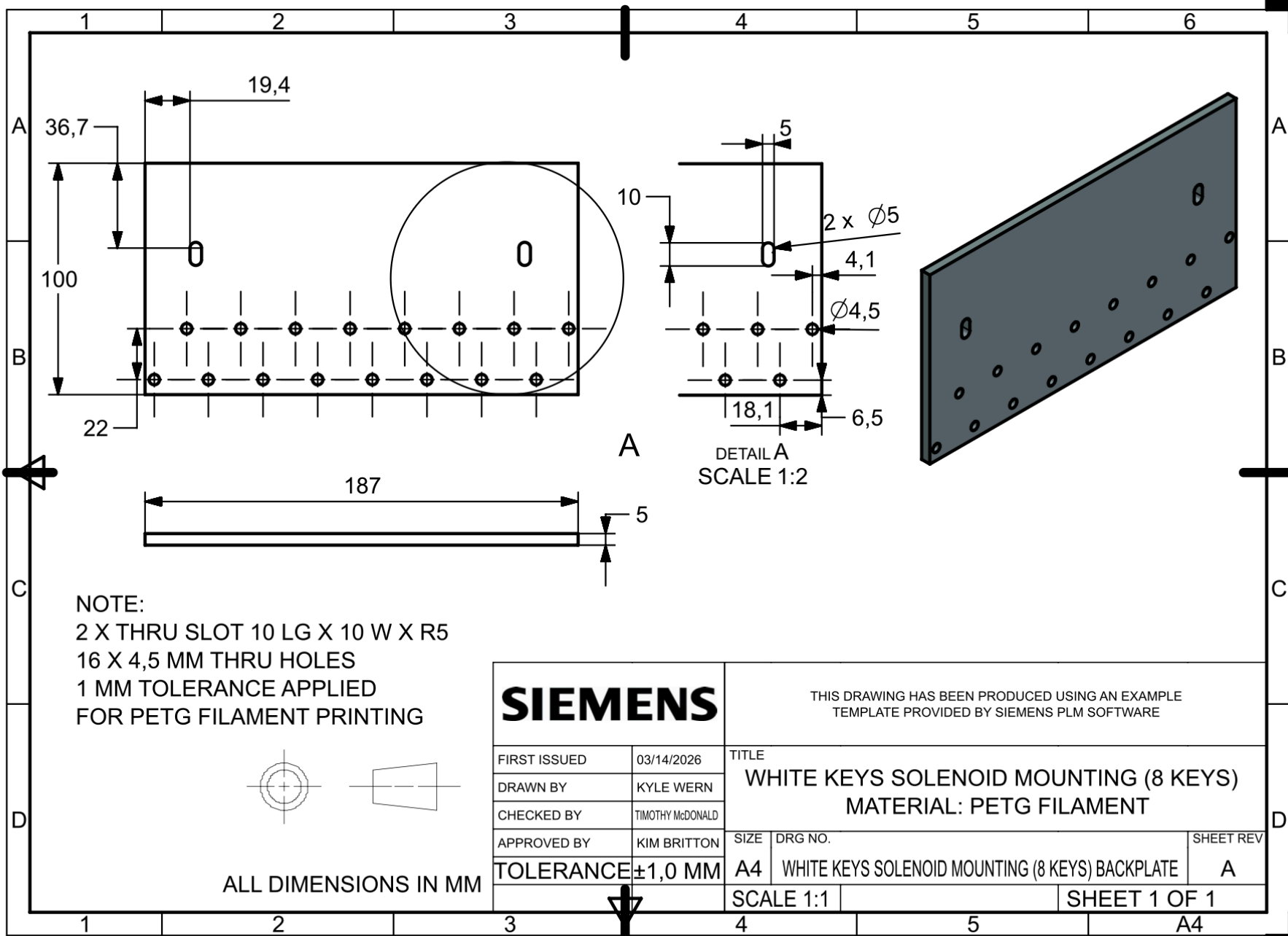
ALL DIMENSIONS IN MM

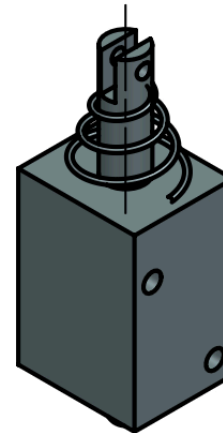
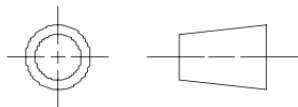
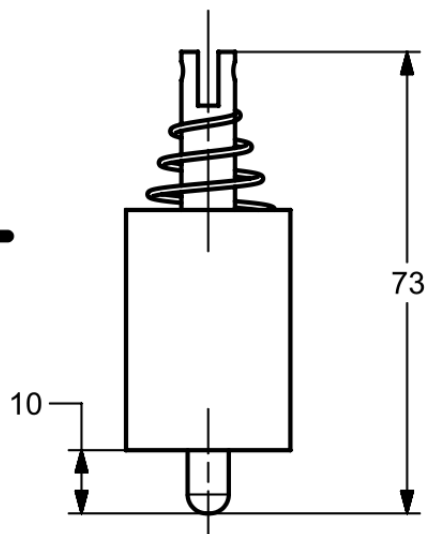
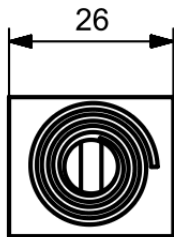
SIEMENS

FIRST ISSUED	03/14/2026
DRAWN BY	KYLE WERN
CHECKED BY	TIMOTHY McDONALD
APPROVED BY	KIM BRITTON
TOLERANCE	±1,0 MM

THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE			
TITLE			
WHITE KEYS OCTAVE ASSEMBLY (8 KEYS)			
MATERIAL: MISC.			
SIZE	DRG NO.	SHEET REV	
A3	WHITE KEYS OCTAVE ASSEMBLY (8)	A	
SCALE 1:1		SHEET 1 OF 1	

A3





NOTE:
25 N SOLENOID MODELED FROM EXISTING AMAZON PRODUCT:
25 N PUSH PULL TYPE SOLENOID (ASIN: B07Z4JQPN)

SIEMENS

THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE

FIRST ISSUED	03/14/2026
DRAWN BY	KYLE WERN
CHECKED BY	TIMOTHY McDONALD
APPROVED BY	KIM BRITTON
TOLERANCE	N/A

TITLE 25 N PUSH PULL TYPE SOLENOID (ASIN: B07Z4JQPN) MATERIAL: MISC.			
SIZE	DRG NO.	SHEET REV	
A4	25 N SOLENOID	A	
SCALE 1:1		SHEET 1 OF 1	

ALL DIMENSIONS IN MM

A4

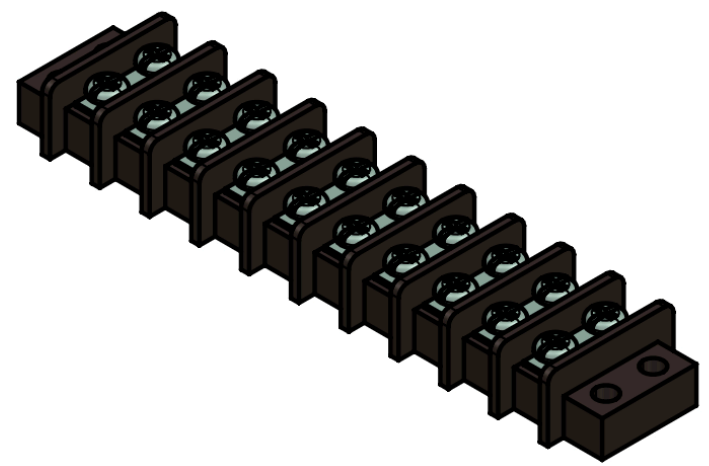
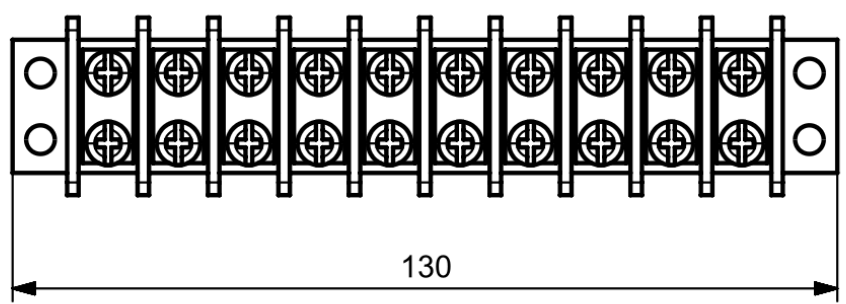
1 2 3 4 5 6

A

B

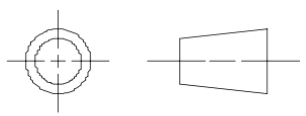
C

D



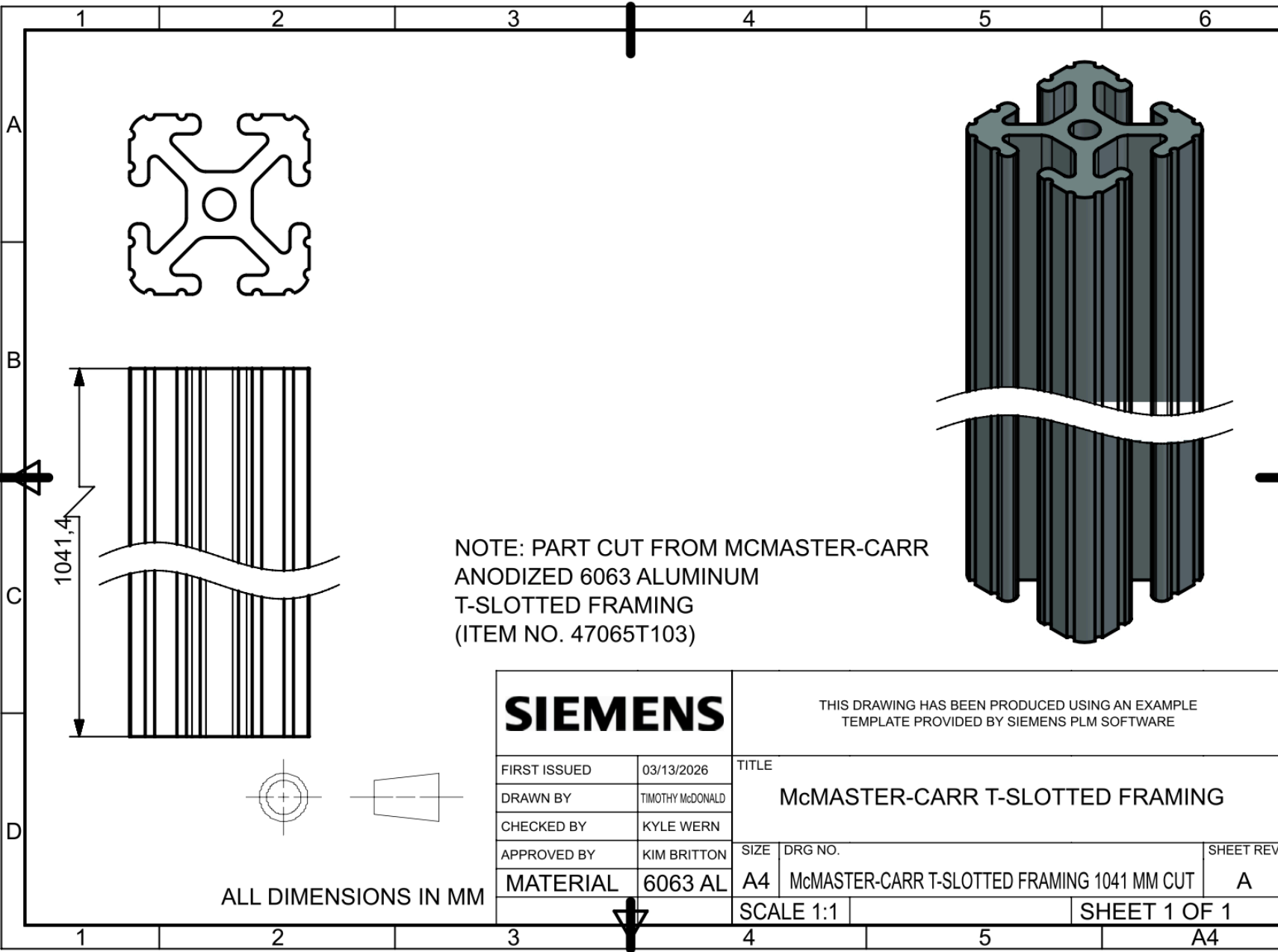
NOTE:
BARRIER BLOCK FOUND ON GRABCAD:
BARRIER BLOCK TE 1-1546670-0
BY DAVID HENDERSON (PUBLISHED 09/08/2025)

SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	09/08/2025	TITLE BARRIER BLOCK TE 1-1546670-0 MATERIAL: MISC.		
DRAWN BY	DAVID HENDERSON			
CHECKED BY	N/A			
APPROVED BY	N/A	SIZE	DRG NO.	SHEET REV
TOLERANCE	N/A	A4	BARRIER BLOCK TE 1-1546670-0	A
		SCALE 1:1	SHEET 1 OF 1	



ALL DIMENSIONS IN MM

1 2 3 4 5 A4



NOTE: PART CUT FROM MCMaster-CARR
ANODIZED 6063 ALUMINUM
T-SLOTTED FRAMING
(ITEM NO. 47065T103)

SIEMENS

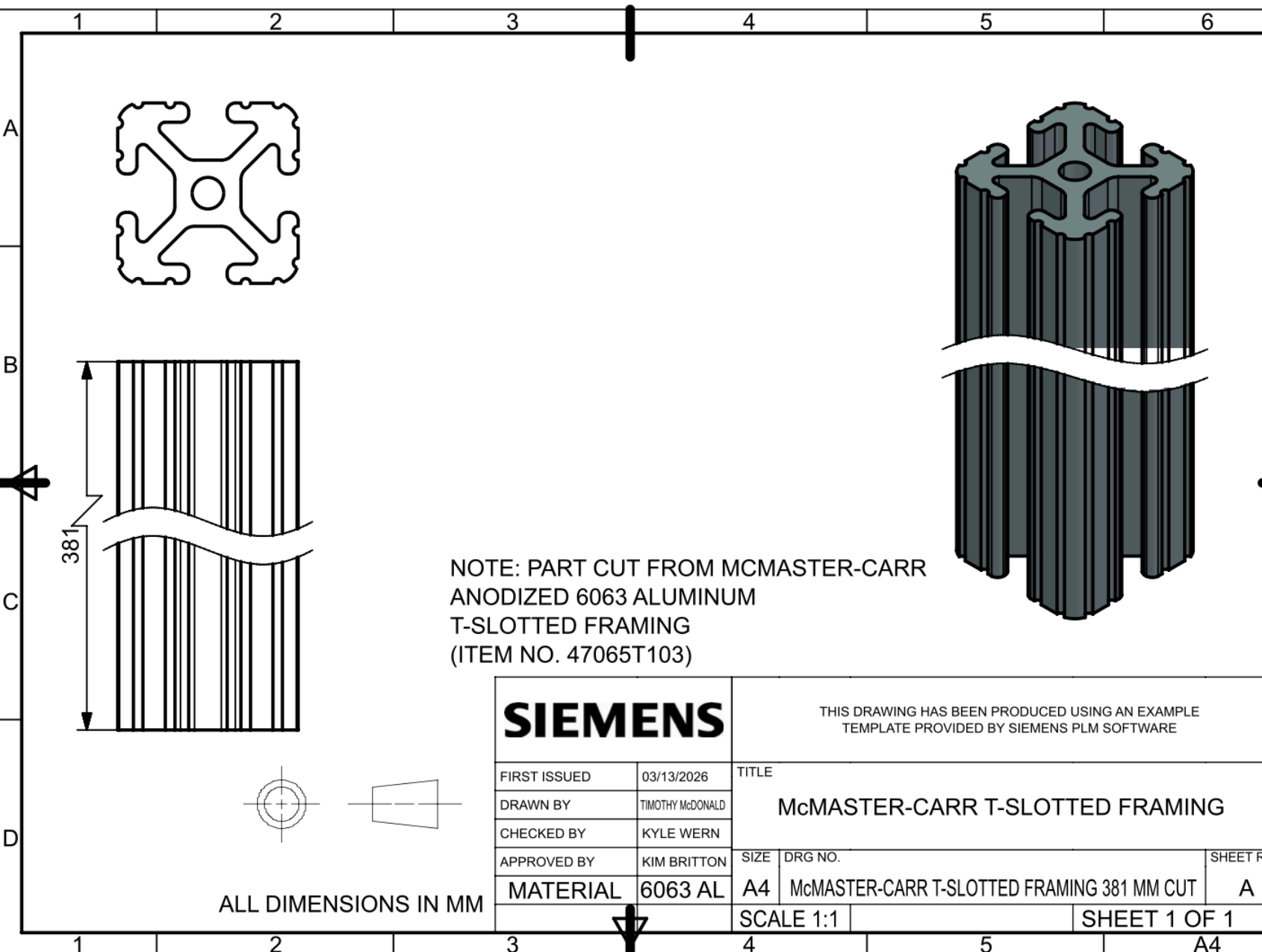
THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE

FIRST ISSUED	03/13/2026
DRAWN BY	TIMOTHY McDONALD
CHECKED BY	KYLE WERN
APPROVED BY	KIM BRITTON
MATERIAL	6063 AL

TITLE			
McMASTER-CARR T-SLOTTED FRAMING			
SIZE	DRG NO.	SHEET REV	
A4	McMASTER-CARR T-SLOTTED FRAMING 1041 MM CUT	A	

ALL DIMENSIONS IN MM

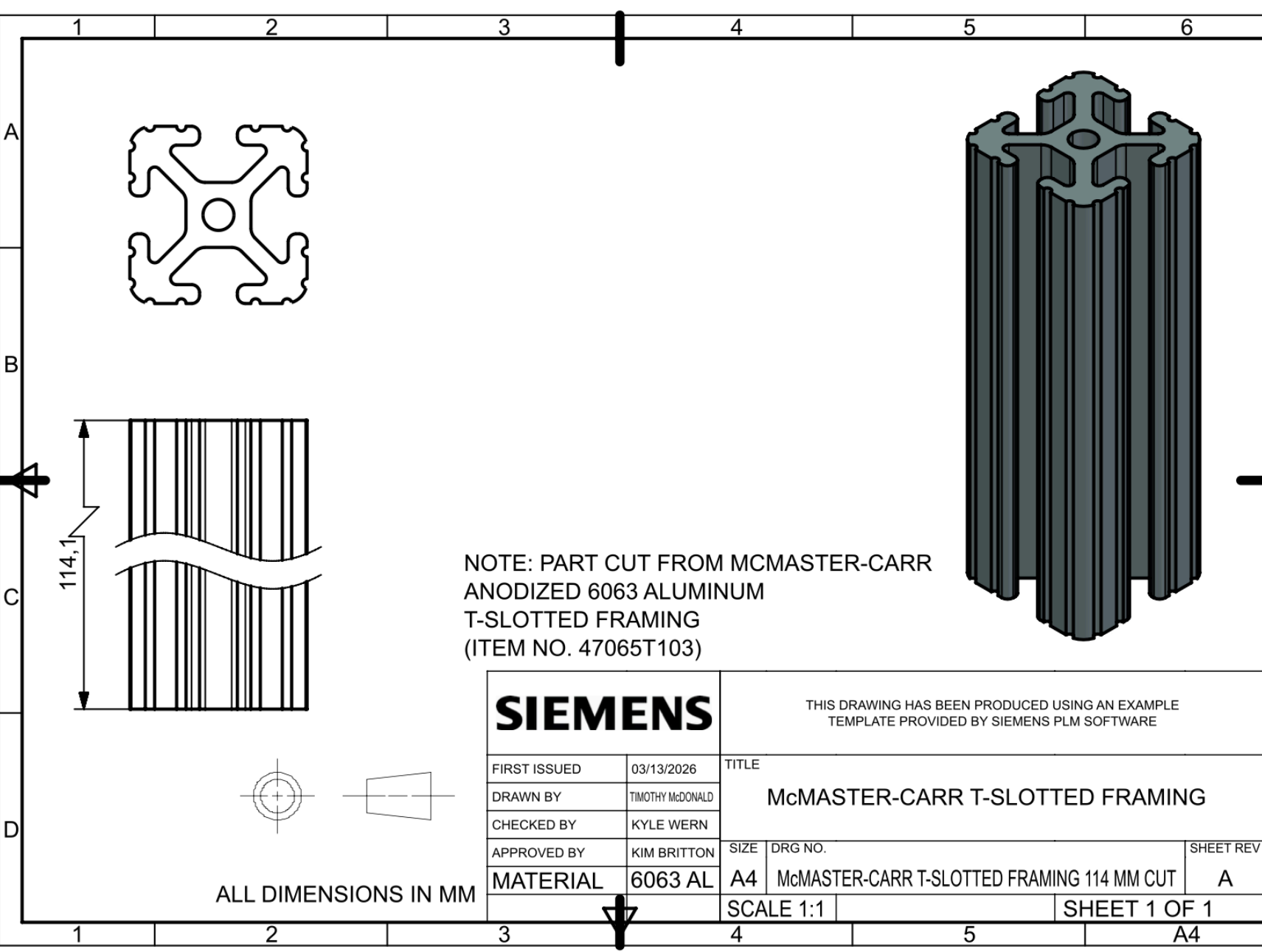
SCALE 1:1	SHEET 1 OF 1
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NOTE: PART CUT FROM MCMaster-CARR
ANODIZED 6063 ALUMINUM
T-SLOTTED FRAMING
(ITEM NO. 47065T103)

SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE			
FIRST ISSUED	03/13/2026	TITLE McMASTER-CARR T-SLOTTED FRAMING			
DRAWN BY	TIMOTHY McDONALD				
CHECKED BY	KYLE WERN				
APPROVED BY	KIM BRITTON	SIZE	DRG NO.	SHEET REV	
MATERIAL	6063 AL	A4	McMASTER-CARR T-SLOTTED FRAMING 381 MM CUT	A	
		SCALE 1:1		SHEET 1 OF 1	
				A4	

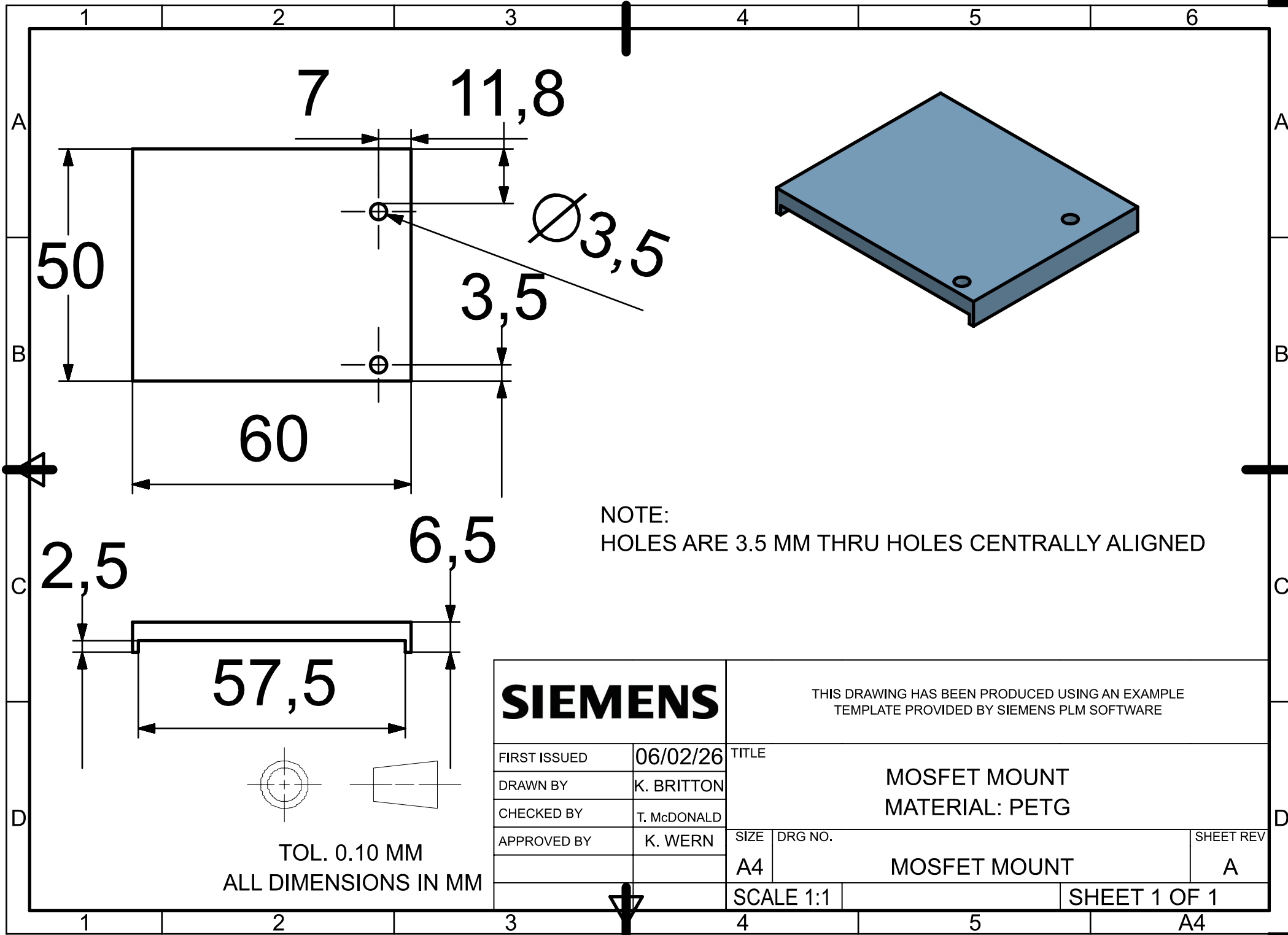
ALL DIMENSIONS IN MM

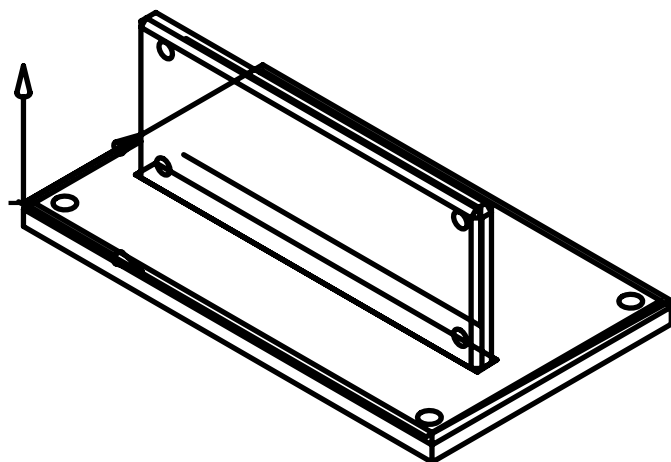
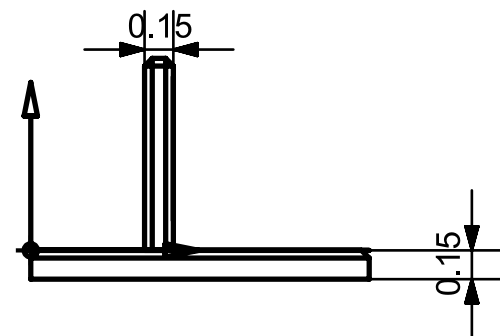
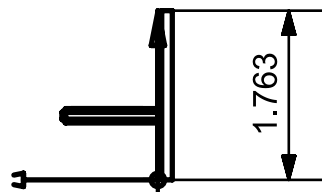
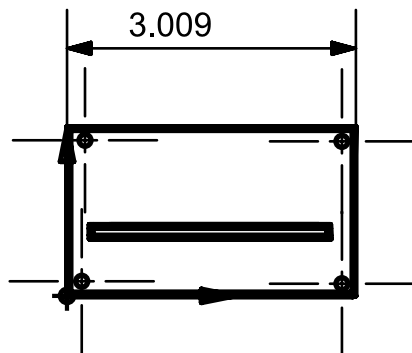


NOTE: PART CUT FROM MCMaster-CARR
ANODIZED 6063 ALUMINUM
T-SLOTTED FRAMING
(ITEM NO. 47065T103)

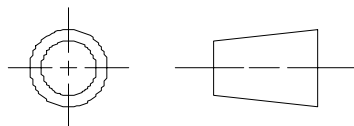
SIEMENS		THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE		
FIRST ISSUED	03/13/2026	TITLE McMASTER-CARR T-SLOTTED FRAMING		
DRAWN BY	TIMOTHY McDONALD			
CHECKED BY	KYLE WERN			
APPROVED BY	KIM BRITTON	SIZE	DRG NO.	SHEET REV
MATERIAL	6063 AL	A4	McMASTER-CARR T-SLOTTED FRAMING 114 MM CUT	A
		SCALE 1:1	SHEET 1 OF 1	

ALL DIMENSIONS IN MM





ALL DIMENSIONS IN INCHES
 MATERIAL: PETG
 TOLERANCE: +/-0.04 IN (1.0 MM) U.N.O.
 3D PRINTED COMPONENT



ALL DIMENSIONS IN INCHES

SIEMENS

THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
 TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE

FIRST ISSUED	06/03/2026
DRAWN BY	D. MARTIN
CHECKED BY	T. MCDONALD
APPROVED BY	K. WERN

TITLE		
PCA9685 MOUNT		
SIZE	DRG NO.	SHEET REV
A	capstonePCAmount1 (3)	A
SCALE 1:1		SHEET 1 OF 1